College of Engineering – Department of Electronics Engineering

CURRICULUM

Bachelor of Science in Aerospace Engineering

Academic Year 2026-2027

Reference CMOs: CMO No. 86, S. 2017, CMO No. 20, s. 2013, CMO No. 39, s. 2021, and CMO 40, s. 2021

Curriculum Description

The Bachelor of Science in Aerospace Engineering program is designed to cultivate engineers with a robust foundation in mathematics, physics, and engineering principles, enabling them to design, analyze, and innovate within the aerospace sector. Emphasizing both aeronautical and astronautical disciplines, the program prepares students to tackle complex challenges in aircraft and spacecraft systems, propulsion, structures, and control systems.

Students are encouraged to develop a systems engineering perspective, fostering the ability to integrate various subsystems into cohesive, efficient, and reliable aerospace vehicles. Through hands-on projects, internships, and research opportunities, graduates gain practical experience, preparing them for careers in aerospace industries, research institutions, and governmental agencies.

The program also instills a commitment to lifelong learning, ethical responsibility, and global collaboration, recognizing the interdisciplinary and international nature of modern aerospace endeavors. By aligning with current industry standards and technological advancements, the curriculum ensures that graduates are well-equipped to contribute meaningfully to the advancement of aerospace engineering.

Program Education Objectives

The aerospace engineering alumni three to five years after graduation shall:
PEO 1: Practice as a specialist with competency in the design, development, and analysis of aerospace systems while adhering to ethical, environmental, and societal standards.
PEO 2: Exhibit leadership, communication, and teamwork skills necessary to function productively in diverse, multidisciplinary, and multicultural work environments.
PEO 3: Pursue continuous professional growth through licensure, graduate education, certifications, research, or self-directed learning in engineering and related fields.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex aerospace engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

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SO5 Problem Analysis. Identify, formulate, and solve complex aerospace engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex aerospace engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex aerospace engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Advanced Engineering Mathematics for AeE	3	0	3
Subtotal	**17**	**0**	**17**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Subtotal	**6**	**6**	**8**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Computer Programming	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Drafting and Design I	0	3	1
Computer-Aided Drafting and Design II	0	3	1
Basic Occupational Safety and Health	3	0	3
Technopreneurship	3	0	3
Environmental Science and Engineering	3	0	3
Statics of Rigid Bodies	3	0	3

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Dynamics of Rigid Bodies	2	0	2
Numerical Methods and Analysis	3	0	3
Mechanics of Deformable Bodies	3	0	3
Professional Orientation	2	0	2
Subtotal	**22**	**15**	**27**
D. Allied Courses			
Basic Electrical Engineering	2	3	3
Electronic Circuits: Devices and Analysis	3	3	4
Principles of Communication Systems	3	3	4
Thermodynamics	3	0	3
Fluid Dynamics	3	0	3
Feedback and Control System	3	3	4
Subtotal	**17**	**12**	**21**
E. Professional Courses			
Aerospace Engineering Orientation	2	0	2
Aerospace Flight Mechanics	3	0	3
Civil Air Laws, Regulations and Professional Ethics	2	0	2
Aerospace Instrumentation and Control	3	0	3
Aerodynamics	3	3	4
Aerospace Structures	3	0	3
Propulsion Systems	2	3	3
Orbital Mechanics	3	0	3
Aerospace Materials and Manufacturing	2	3	3
Astrodynamics	3	0	2
Spacecraft Systems Engineering	3	0	3
Aerospace Vehicle Stability & Control	3	0	3
Supply Chain Management for AeE	3	0	3
AEE Capstone Design I	0	3	1
AEE Capstone Design II	0	3	1
Subtotal	**35**	**15**	**39**
F. Industry Placement			
Professional Placement 1	640 hours		8
Professional Placement 2	640 hours		8
Subtotal	**1280 hours**		**16**
G. Technical Elective			
AeEE Elective 1	2	3	3
AeEE Elective 2	2	3	3
Subtotal	**4**	**6**	**6**
TOTAL TECHNICAL COURSES	**101**	**54**	**134**
NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
The Contemporary World	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3

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Ethics	3	0	3
Subtotal	**24**	**0**	**24**
B. Filipino/Literature/Mandated Courses			
Life and Works of Rizal	3	0	3
Linear Algebra for AI	3	0	3
Machine Learning and Neural Network	2	3	3
Entrepreneurial Mind	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness			
PATHFit 1 - Movement Competency Training	2	0	2
PATHFit 2 - Exercise-based Fitness Activities	2	0	2
PATHFit 3 – Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Traditional and Recreational Games)	2	0	2
PATHFit 4 – Menu of Dance, Sports, Martial Arts, Group Exercise, Outdoor Adventure Activities Menu Offering: (Team Sports, Basketball and Volleyball)	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
E. Institutional Mandated Course			
Life and Works of Rizal	3	0	3
Subtotal	**3**	**0**	**3**
TOTAL NON-TECHNICAL COURSES	**52**	**3**	**53**
GRAND TOTAL	**153**	**57**	**187**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	17
B. Natural/Physical Sciences	8
C. Basic Engineering Sciences	27
D. Allied Courses	21
E. Professional Courses	39
F. Industry Placements	16
G. Technical Electives	6
II. NON-TECHNICAL COURSES	
A. General Education Courses	24
B. Filipino/Literature/Mandated Courses	12
C. PATHFit 1-4	8
D. NSTP	6
E. Institutional Mandated Course -Foreign Language	3
GRAND TOTAL	**187**

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CURRICULUM STRUCTURE

FIRST YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
CHEM 101	Chemistry for Engineers	4	3	3		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 106	Purposive Communication	3	3	0		
GEd 101	Understanding the Self	3	3	0		
MATH 101	Differential Calculus	4	4	0		
NSTP 111	National Service Training Program 1	3	3	0		
PATHFit 1	Movement Competency Training	2	2	0		
ENGG 102	Engineering Drawing	1	0	3		
ENGG 101	Introduction to Engineering***	1	0	3		
	TOTAL	**24**	**21**	**9**		
FIRST YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
PHYS 101	Physics for Engineers	4	3	3	MATH 101	
GEd 108	Art Appreciation	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 104	The Contemporary World	3	3	0		
MATH 102	Integral Calculus	4	4	0	MATH 101	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
CpE 101	Computer Programming	1	0	3		
GEd 109	Science, Technology and Society***	3	3	0		
	TOTAL	**26**	**24**	**6**		

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FIRST YEAR						
Midterm						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
GEd 107	Ethics	3	3	0		
ME 101	Thermodynamics	3	3	0	PHYS 101	
IMC 101	Foreign Language	*3*	*3*	0		
	TOTAL	***9***	***9***	**0**		
SECOND YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
AEE 101	Aerospace Engineering Orientation***	2	2	0		
AEE 102	Advanced Engineering Mathematics for AeE	3	3	0		
ECE 405	Electronics Circuits: Devices and Analysis	4	3	3	PHYS 101	EE 205
EE 205	Basic Electrical Engineering	3	2	3	PHYS 101	
ME 206	Fluid Mechanics	3	3	0	ME 101	ENGG 107
MATH 201	Differential Equations	3	3	0	MATH 402	
ENGG 107	Statics of Rigid Bodies	3	3	0		
ENGG 121	Computer-Aided Drafting and Design I	1	0	3	ENGG 402	
AI 101	Linear Algebra for Artificial Intelligence	3	3	0		
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 1	2	0	2	PATHFit 1 & 2	
	Total	27	24	9		
SECOND YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
ENGG 108	Dynamics of Rigid Bodies	2	2	0	ENGG 107	
ENGG 115	Numerical Methods and Analysis	3	3	0	AEE 102, CPE 101	
ENGG 118	Mechanics of Deformable Bodies	3	3	0	ENGG 417	ENGG 108

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ENGG 122	Computer-Aided Drafting and Design II	1	0	3	ENGG 121	
AEE 103	Aerospace Flight Mechanics	3	3	0	AEE 101	ME 206
AEE 104	Civil Air Laws, Regulations and Professional Ethics***	2	2	0	AEE 101	
ECE 302	Principles of Communication Systems	4	3	3	ECE 405	
AI 102	Probability and Statistics for AI	3	3	0	AI 101	
ENGG 125	Professional Orientation	2	2	0		
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor & Adventure Activities 2	2	2	0	PATHFit 2	
	Total	25	23	6		
THIRD YEAR						
First Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
ENGG 126	Professional Placement 1	8	640 hours		ENGG 125	
	Total	8	640 hours			
THIRD YEAR						
Second Semester						
Course Code	Course Title	**Credit Units**	**No of Hr./s**		Pre-requisite/s	Co-requisite/s
			Lec	**Lab**		
AEE 301	Aerospace Instrumentation and Control	3	3	0	AEE 102, EE 205	
AEE 302	Aerodynamics	4	3	3	ME 101, ME 206, AEE 103	
AEE 303	Aerospace Structures***	3	3	0	AEE 103	
AEE 304	Propulsion Systems	3	2	3	AEE 103, ENGG 118	
AEE 310	Capstone Design for Aerospace Engineering I	1	0	3	ENGG 126	
AeEE 301	Aerospace Elective I	3	2	3	3rd yr standing	
AI 103	Machine Learning and Deep Learning Applications	3	2	3	AI 102, CPE 401	
ECE 303	Feedback and Control System	4	3	3	AEE 102, EE 205	
	Total	26	18	18		
THIRD YEAR						
Midterm Semester						

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Course Code	Course Title	Credit Units	No of Hr./s		Pre-requisite/s	Co-requisite/s
			Lec	Lab		
ENGG 116	Environmental Science and Engineering	3	3	0	CHEM 101	
GEdE 103	The Entrepreneurial Mind	3	3	0		
ENGG 114	Basic Occupational Safety and Health	3	3	0		
	TOTAL	**9**	**9**	**0**		
FOURTH YEAR						
First Semester						
Course Code	Course Title	Credit Units	No of Hr./s		Pre-requisite/s	Co-requisite/s
			Lec	Lab		
ENGG 105	Technopreneureship	3	3	0	4th yr Standing	
AEE 401	Orbital Mechanics	3	3	0	AEE 302, AEE 103	
AeEE 401	AeE Elective II	3	2	3	AeEE 301	
AEE 402	Aerospace Materials and Manufacturing***	3	2	3	4th yr Standing	
AEE 403	Astrodynamics	2	3	0	4th yr standing	
AEE 311	Capstone Design for Aerospace Engineering II	1	0	3	AEE 310	
AEE 404	Spacecraft Systems Engineering	3	3	0	4th yr standing	
AEE 405	Aerospace Vehicle Stability & Control	3	3	0	AEE 103, AEE 302	
GEd 103	Life and Works of Rizal	3	3	0		
AEE 406	Supply Chain Management for AeE	3	3	0	4th yr standing	
	Total	27	25	9		
FOURTH YEAR						
Second Semester						
Course Code	Course Title	Credit Units	No of Hr./s		Pre-requisite/s	Co-requisite/s
			Lec	Lab		
IP	Industry Placement 2	8	640 hours			
GRAND TOTAL		187	153	57		

**The following courses will be co-delivered in collaboration with industry partners to provide students with practical insights, real-world applications, and exposure to current industry standards and practices.*

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INDUSTRY BASED LEARNING

INDUSTRY -BASED CURRICULUM FOR BS AEROSPACE ENGINEERING			
YEAR	**Semester 1**	**Semester 2**	**Midterm**
1	Academic	Academic	Academic
2	Academic	Academic	
3	Industry Placement	Academic	Academic
4	Academic	Industry Placement	**GRADUATE**